

HP M1005 MFP

A value product that delivers

The HP M1005 MFP is a printer, scanner, and copier targeted at the entry-level consumer or small SoHo. It is pretty sleek and doesn't take up too much desktop space—just 43.7 x 36.3 cm. It looks good, too, and weighs a manageable 8.5 kg.

The control panel is intuitive with a two-line, 16-character display, but the menu offers a limited set of functions to control. Input tray capacity is 150 sheets, which is fairly good for its class. Output tray capacity is 100—more than sufficient for this category of MFDs. Unlike with many HP drivers, the installation for this one was fast. One Windows restart later, the M1005 was ready to start multi-functioning...



Print resolution is the standard 600 x 600 dpi, and 32 MB of buffer is good enough to handle long print queues. It took only 14.2 seconds to warm up and start printing. The first text print took just under 10 seconds, while the combi-print required 12.8

seconds. Maximum print speed is around 13 ppm for all kinds of documents. The quality of the prints is not much to boast about, especially the graphics. Inverse text, though, is printed better than on more expensive machines.

The flatbed scanner scans documents of up to A4 size at 1200 x 1200 dpi in 24-bit colour. It took us just 7.3 seconds for the monochrome scan, but it got lethargic during the photo scan, taking over a minute and a half. Scan quality is excellent, though—colour and detail capture was superior to that of most of the MFDs we've tested thus far! This was the *only* MFD here that was perfect in the OCR test.

The first photocopy took 12.8 seconds, but the quality was below par—a little washed out.

HP does not bundle their software suite with this one: it is a "value" product. But they do supply the impressive Readiris Pro OCR software for PC as well as Mac. The monthly duty cycle of the M1005 is 5,000 prints; understandable, since this is an entry-level MFD. The cartridge has a duty cycle of 2,000 prints and costs Rs 2,961.

Rs 12,999 makes the M1005 an attractive choice if your work won't be hampered by the relatively slow printing speeds and the lack of standalone fax.

How We Tested

The Test Rig

For all the tests, we used 100 gsm (grams per square metre) A4-size paper from Berga.

Test Rig

Processor	Intel Core 2 Extreme X6800 @ 2133 MHz
Motherboard	Gigabyte GA-965G-DS3
Chipset	Intel 965G
RAM	1 GB Kingston HyperX DDR2 @ 533 MHz
Graphics Card	NVIDIA GeForce 7300 GS
Host hard drive	Seagate Barracuda 750 GB SATA II
OS	Windows XP with SP2

Features

We noted features such as printer and scanner resolutions, scanner type and size, presence of a fax unit and ADF (Automatic Document Feeder), input tray capacity, and more. For the copier tests, we noted the MFDs' ability to work as standalone devices, the maximum number of pages that can be printed in multi-copy mode, the facility to reduce or enlarge a copy, and more.

Warm-up Time

All laser MFDs need a "warm up" when they are first started, only after which they are ready to function. We noted how long each MFD took to warm up.

The Printer Tests

We used a text document to test the raw print speeds of the printers on the MFDs. To test the ability of the MFDs to handle the various aspects of a regular office document (which can have both text and graphics), we created a "combi-document" that has black text interspersed with graphs and images. We used the Draft and best quality settings for the text and combi-document tests. We gauged the speeds by sending the Print command and clocking the time the MFDs took to print the first sheet.

The Scanner Tests

We tested the scanners for speed using an average of five previews of an A4-size image. We did this with the scanner just turned on, so as to incorporate the warm-up time. We imported the same A4-size image at 150 and 600 dpi in Adobe Photoshop CS2, and scanned a full-text document in B&W mode at 200 dpi. We then put the scanners through a test to determine colour differentiating capabilities using the Kodak IT8 card (which has different shades of colour in a gradient). Next, we scanned the resolution chart at 300 dpi—this has five boxes consisting of parallel lines very close to each other, with the boxes arranged in order of density of the lines. This test was, of course, to check whether the scanners could scan and differentiate between the parallel lines, and gave us a measure of the scanners' ability to capture fine detail. Finally, we did an OCR test to check the scanner's ability to differentiate between light and dark areas: we scanned a print article to the trial version of ABBYY Fine Reader

Pro 8.0 PE, and counted the wrongly-interpreted letters.

The Copier Tests

We copied the colour combi-document to black-and-white and rated the MFDs on speed and print quality. Here, too, we noted the time it took for the first print, and then noted the number of pages printed in the minute after that to calculate the maximum achievable copies per minute (cpm) while multi-copying.

